

Roll No.

TPE-009

B. TECH. (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022
NEURAL NETWORKS AND MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss the biological neural network and artificial neural networks.
Explain the different types of activation functions. 10 Marks (CO2)

OR

- (b) Differentiate between linear regression and logistic regression in detail.
10 Marks (CO2)

2. (a) Explain Hebb's rule in detail with proper equations. How the artificial neural network learns using associative memory ? 10 Marks (CO1)

OR

- (b) A neuron possesses the following properties : 10 Marks (CO1)

(i) Input = 3.0

(ii) Bias = - 2

(iii) Weight = 2.5

Find the neuron output if the transfer functions are hard limit and linear, respectively.

P. T. O.

(2)

3. (a) What are the learning types in the neural networks ? Explain, how neural network performs pattern classification task. 10 Marks (CO1)

OR

- (b) A two input neuron have the following parameters : 10 Marks (CO1)

$B = 1.5$, $W = [2 \ 3]$ and $p = [-4 \ 7]^T$. Calculate the neuron output for the following transfer functions :

- (i) Log sigmoid
- (ii) Saturating linear
- (iii) Hard Limit

4. (a) Discuss the following terms with respect to Artificial Neural Networks :

- (i) Classification
- (ii) Regression
- (iii) Clustering

10 Marks (CO1)

OR

- (b) Why is the tradeoff between bias and variance necessary ?

10 Marks (CO1)

5. (a) Discuss in detail about the two input neuron model and derive its boundary conditions. 10 Marks (CO2)

OR

- (b) What is curve fitting ? How a straight line can be fitted to a given dataset ? 10 Marks (CO2)

Roll No.

TEC-801

B. TECH. (ECE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022
COMPUTER ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Draw the basic structure of computer and explain its various functional units. 10 Marks (CO1, CO2)

OR

- (b) Explain 16 bit instruction code for memory reference instructions with two examples. 10 Marks (CO1, CO2)

2. (a) Discuss register reference instructions with 3 examples.

10 Marks (CO2)

OR

- (b) Draw and explain the flowchart for instruction cycle. 10 Marks (CO2)

3. (a) Explain and classify the following instruction on the basis of memory reference, register reference and Input-Output reference instruction :

(i) LDA

P. T. O.

(2)

- (ii) STA
- (iii) CMA
- (iv) INP
- (v) AND

10 Marks (CO1, CO2)

OR

- (b) Explain 4 phases of an instruction cycle.

10 Marks (CO1, CO2)

4. (a) Explain Booth algorithm for multiplication of two unsigned nos with one help of example.

10 Marks (CO2)

OR

- (b) Explain direct and indirect addressing modes with the help of an example.

10 Marks (CO2)

5. (a) Design BCD adder combinational circuit.

10 Marks (CO3)

OR

- (b) Draw a flow chart for interrupt cycle.

10 Marks (CO3)

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10 Marks (CO2)

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OR

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